

Q:11 2-D ideal, Butterworth and Gaussian LPF & Compare the performance

1. Ideal Low-Pass Filter (ILPF)

$$H(u, v) = \begin{cases} 1, & D(u, v) \leq D_0 \\ 0, & D(u, v) > D_0 \end{cases}$$

Frequency Domain: Sharp/Abrupt cutoff at radius D_0 .

Spatial Domain: Inverse transform is a **Sinc function**.

Effect: Causes severe **Ringings artifacts** (Gibbs phenomenon) around edges.

2. Butterworth Low-Pass Filter (BLPF)

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}}$$

Frequency Domain: Smooth transition; no abrupt jump.

Order (n): Higher n makes the cutoff sharper (closer to Ideal).

Effect: Very little ringing at low orders (ex: $n=2$).

3. Gaussian Low-Pass Filter (GLPF)

$$H(u, v) = e^{-\frac{D^2(u, v)}{2D_0^2}}$$

Frequency Domain: Bell-shaped smooth curve, no sharp cutoff.

Spatial Domain: Inverse transform is also **Gaussian**.

Effect: Zero ringing artifacts. Provides the most natural blur.

Comparison of Filters for Image Enhancement

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Feature	Ideal LPF	Butterworth LPF	Gaussian LPF
Cutoff behavior	Sharp	Smooth (controlled by n)	Very smooth
Ringing effect	High	Moderate	None
Edge preservation	Poor	Moderate	Good
Smoothness	Low	Medium	High
Control parameter	D_0	D_0, n	D_0
Visual quality	Worst (artifacts)	Balanced	Best

